

Lab 1: Packet analysis at application layer using Wireshark
SCSR1213 Network Communications
Universiti Teknologi Malaysia

Objective:

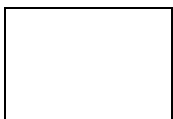
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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Section :__2__



Mark

PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

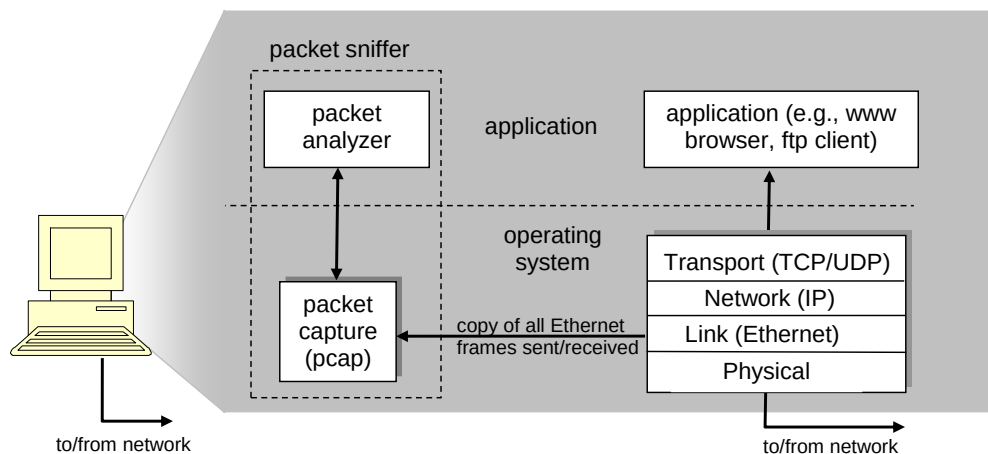


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

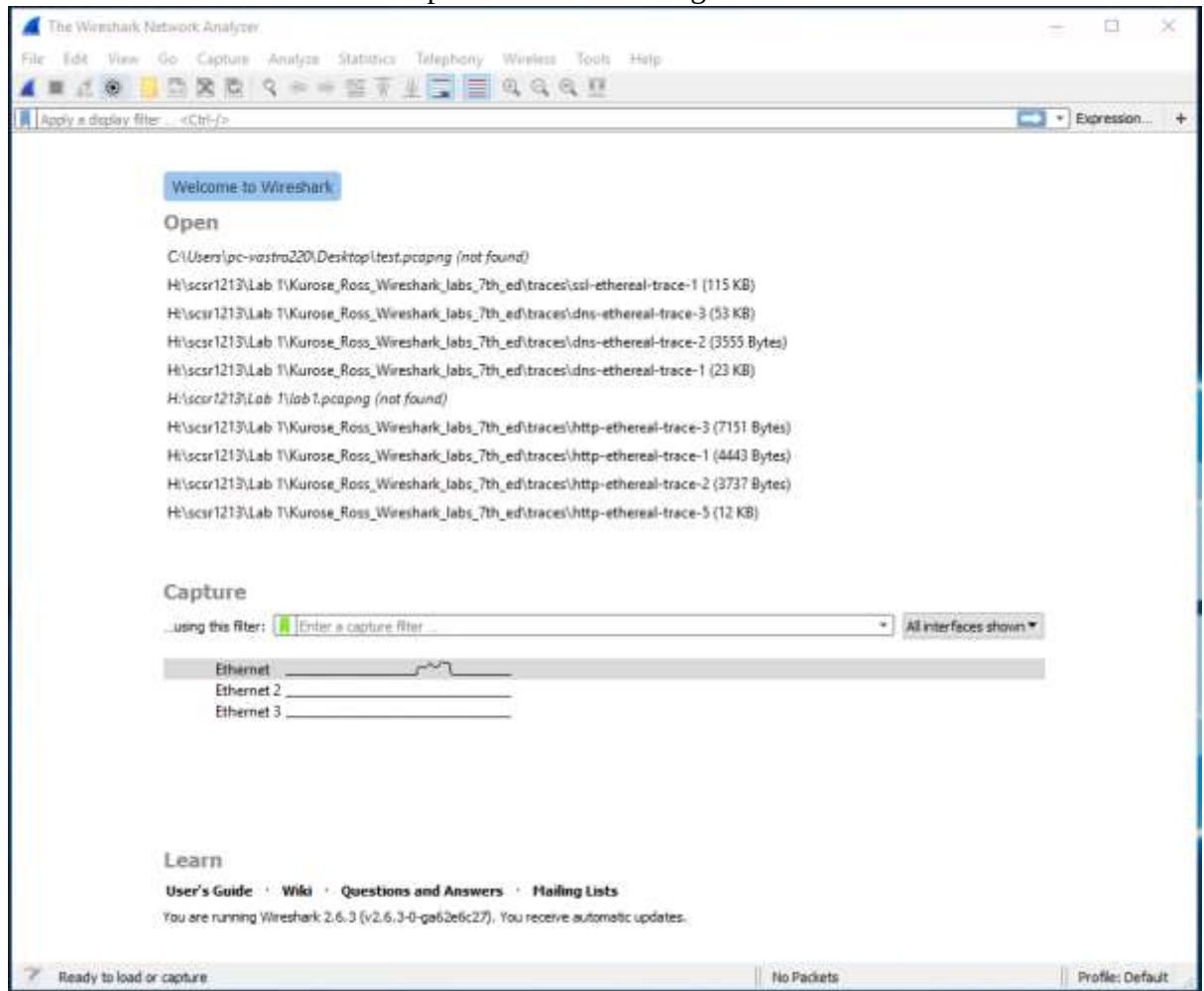


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

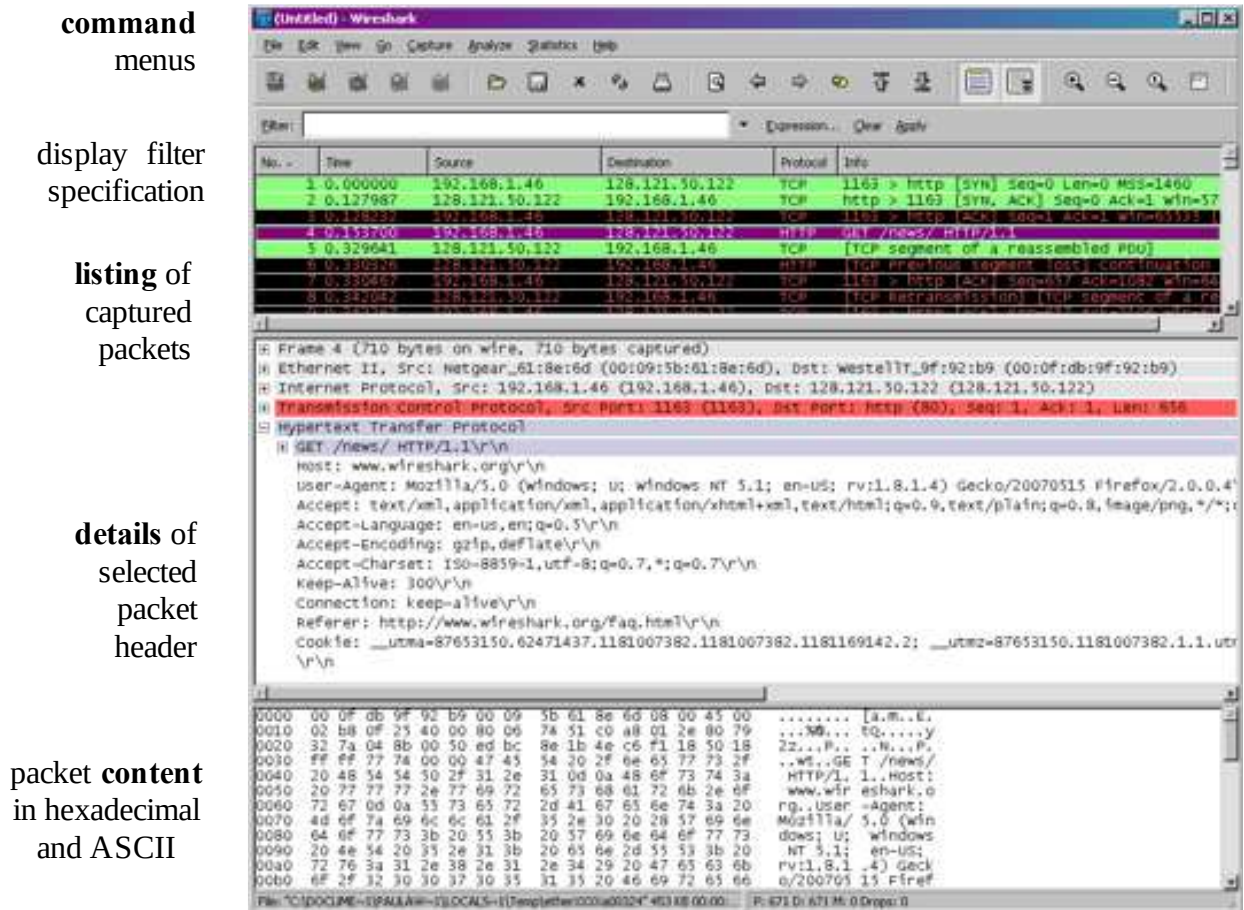


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

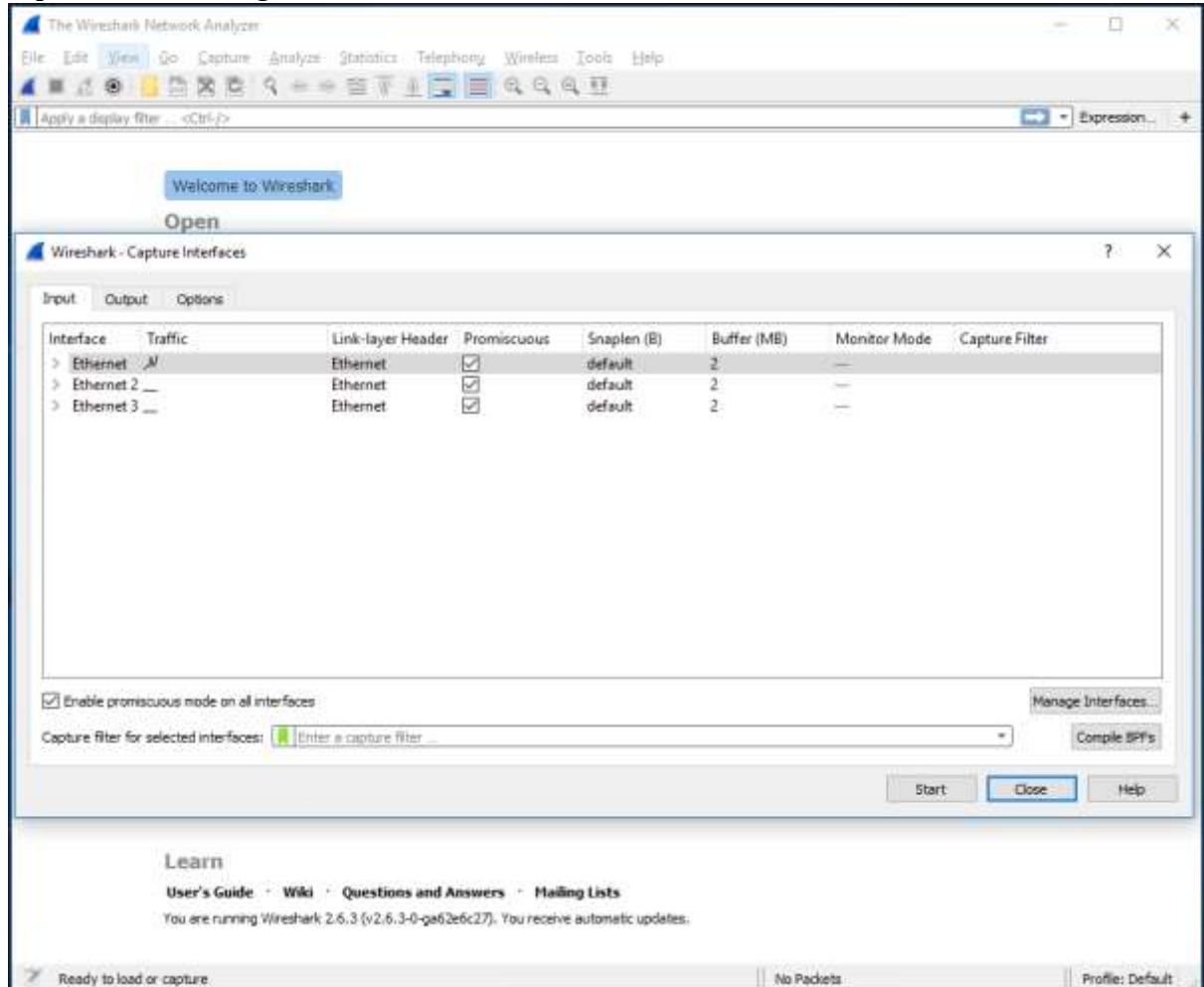


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

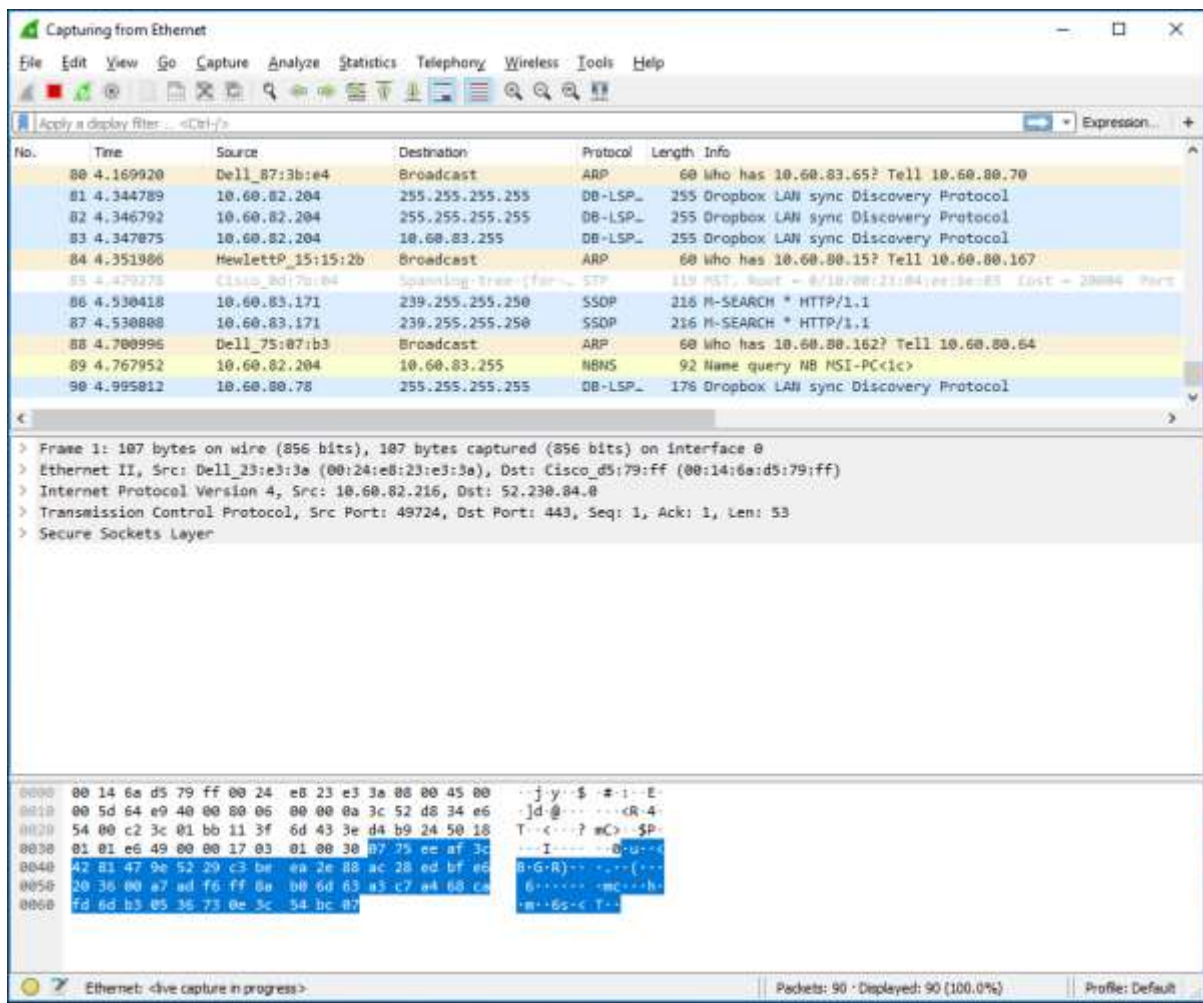


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

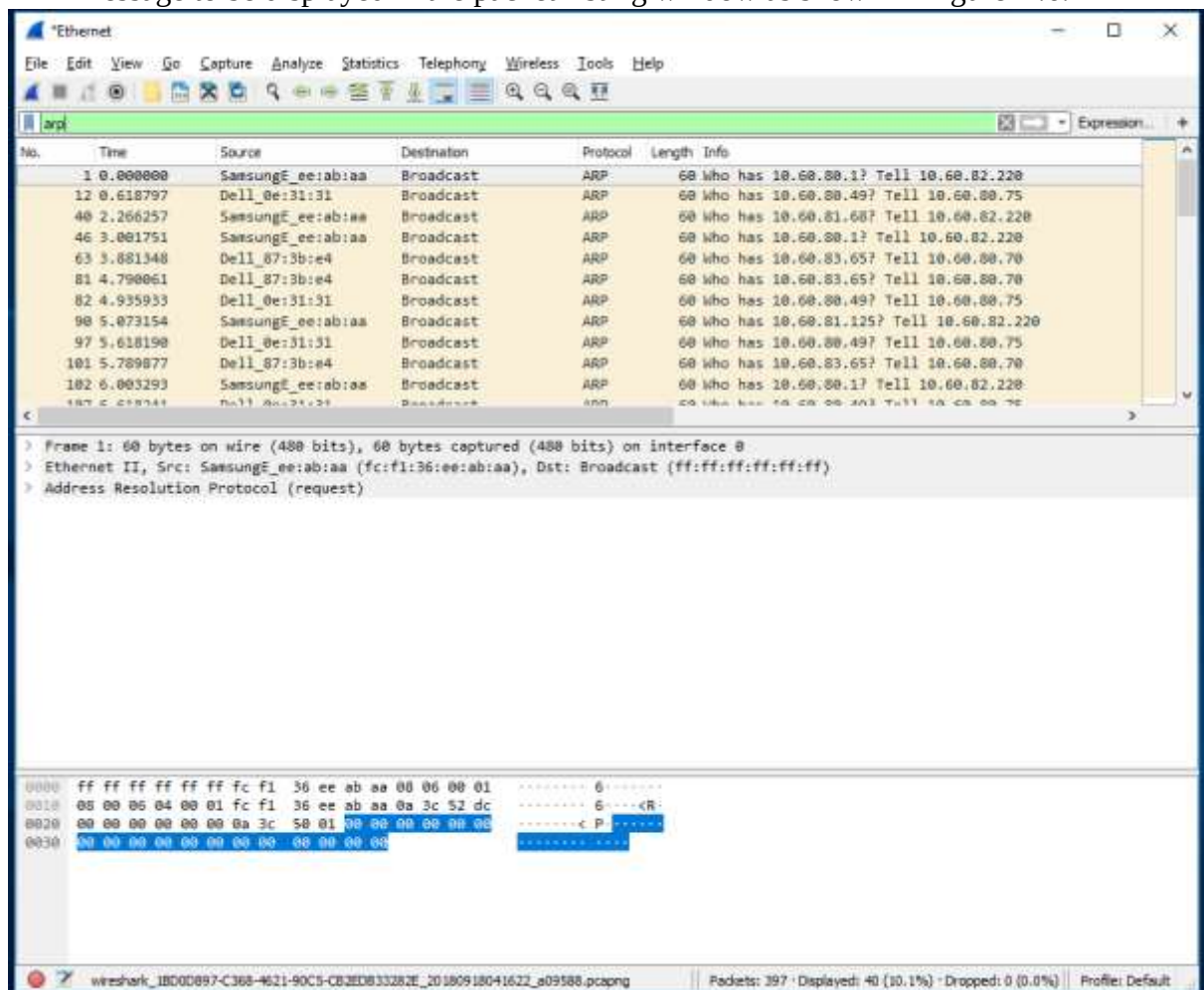


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

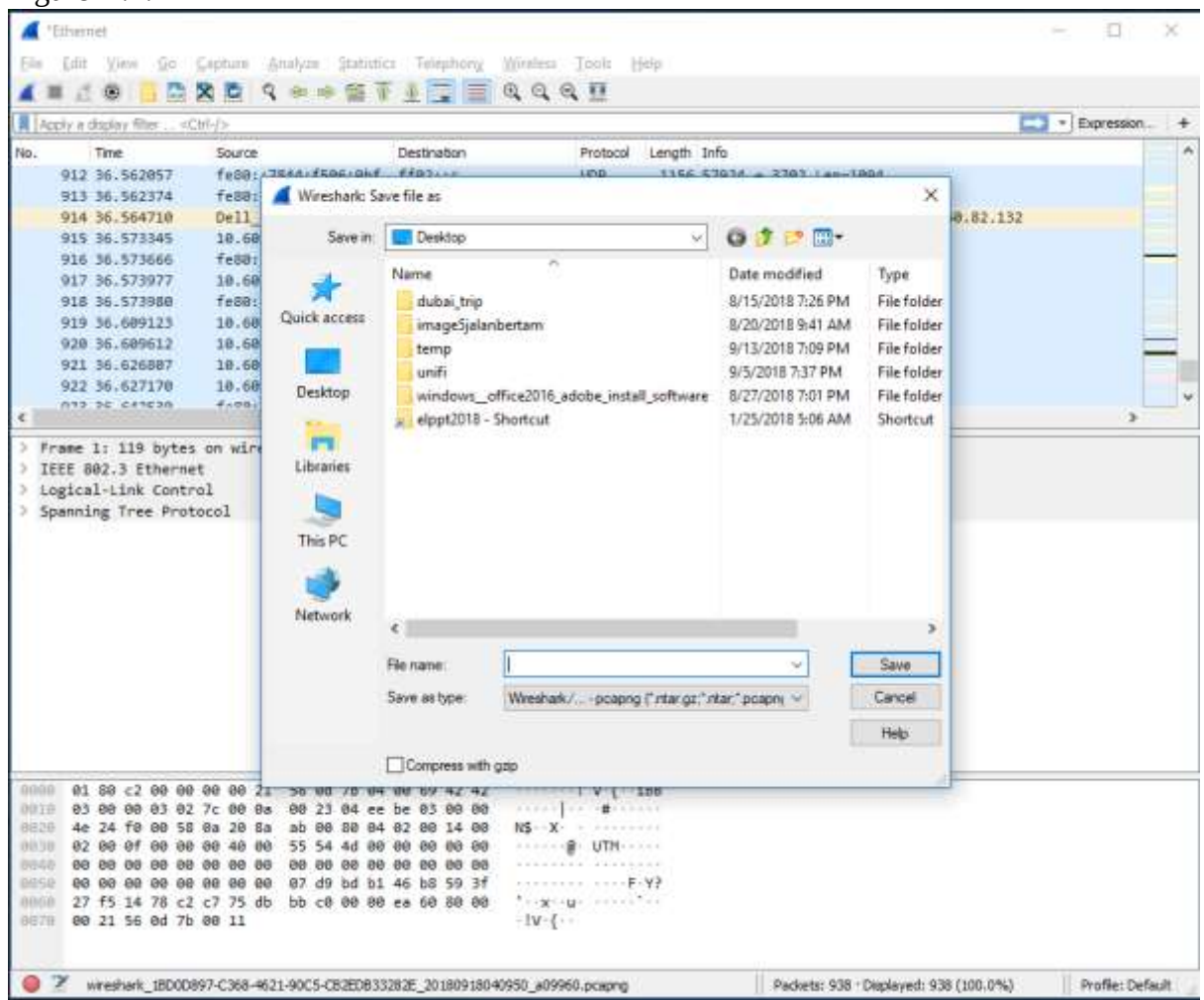


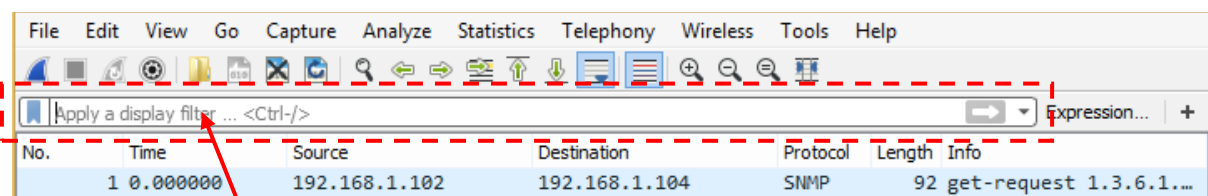
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:

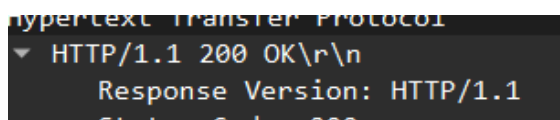


packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?

HTTP version 1.1



2. What is the IP address of the client computer?

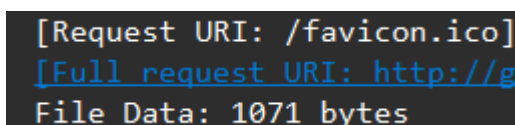
192.168.1.102

3. What is the IP address of the gaia.cs.umass.edu server?

128.119.245.12

4. How many bytes of content are being returned to client browser?

Frame 14: 1071 bytes



Frame 12: 73 bytes

```
[Request URI: /ethereal-labs/lab2-1.html]  
[Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]  
File Data: 73 bytes
```

5. What is the status code returned from the server to client browser?

Frame 12: 200 OK

```
Response Version:  
Status Code: 200
```

Frame 14: 404

```
Status Code: 404
```

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No

```
HTTP/1.1 200 OK\r\n
GET /etherreal-labs/lab2-2.html HTTP/1.1\r\n
Request Method: GET
Request URI: /etherreal-labs/lab2-2.html
Request Version: HTTP/1.1
Host: gaia.cs.umass.edu\r\n
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\n
Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,
Accept-Language: en-us,en;q=0.5\r\n
Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
```

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes, the server explicitly returns the contents of the file. Because the next frame show 200 ok, which means successfully return the contents.

```
HTTP/1.1 200 OK\r\n
Response Version: HTTP/1.1
Status Code: 200
[Status Code Description: OK]
Response Phrase: OK
```

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes. Tue, 23 Sep 2003 05:35:00 GMT\r\n

```
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
If-None-Match: "1bfef-173-8f4ae900"\r\n
Cache-Control: max-age=0\r\n
```

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

304, Not Modified. The server did not explicitly return the contents of the file because the file has no changes.

```
HTTP/1.1 304 Not Modified\r\n
  Response Version: HTTP/1.1
  Status Code: 304
  [Status Code Description: Not Modified]
  Response Phrase: Not Modified
```

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

3

No.	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.268813	128.119.245.12	192.168.1.102	HTTP	1857	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333854	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.580877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

2. To which Internet addresses were these GET requests sent?

Frame 10: 128.119.245.12

Frame 17: 165.193.123.218

Frame 20: 134.241.6.82

No.	Time	Source	Destination	Protocol	Length	Info
10	7.236929	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.268813	128.119.245.12	192.168.1.102	HTTP	1857	HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625	GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609	GET /~kurose/cover.jpg HTTP/1.1
25	7.333854	165.193.123.218	192.168.1.102	HTTP	912	HTTP/1.1 200 OK (GIF89a)
54	7.580877	134.241.6.82	192.168.1.102	HTTP	1096	HTTP/1.0 200 Document follows (JPEG JFIF image)

3. How many bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

3357 bytes

```
Last-modified: Wed, 18 Jun 2003 18:48:24 GMT\r\n
Content-length: 3357\r\n
Accept-ranges: bytes\r\n
Connection: keep-alive\r\n
\r\n
[Request in frame: 17]
[Time since request: 0.027569000 seconds]
[Request URI: /catalog/images/pearson-logo-footer.gif]
[Full request URI: http://www.aw-bc.com/catalog/images/pearson-logo-footer.gif]
File Data: 3357 bytes
```

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

15642 bytes

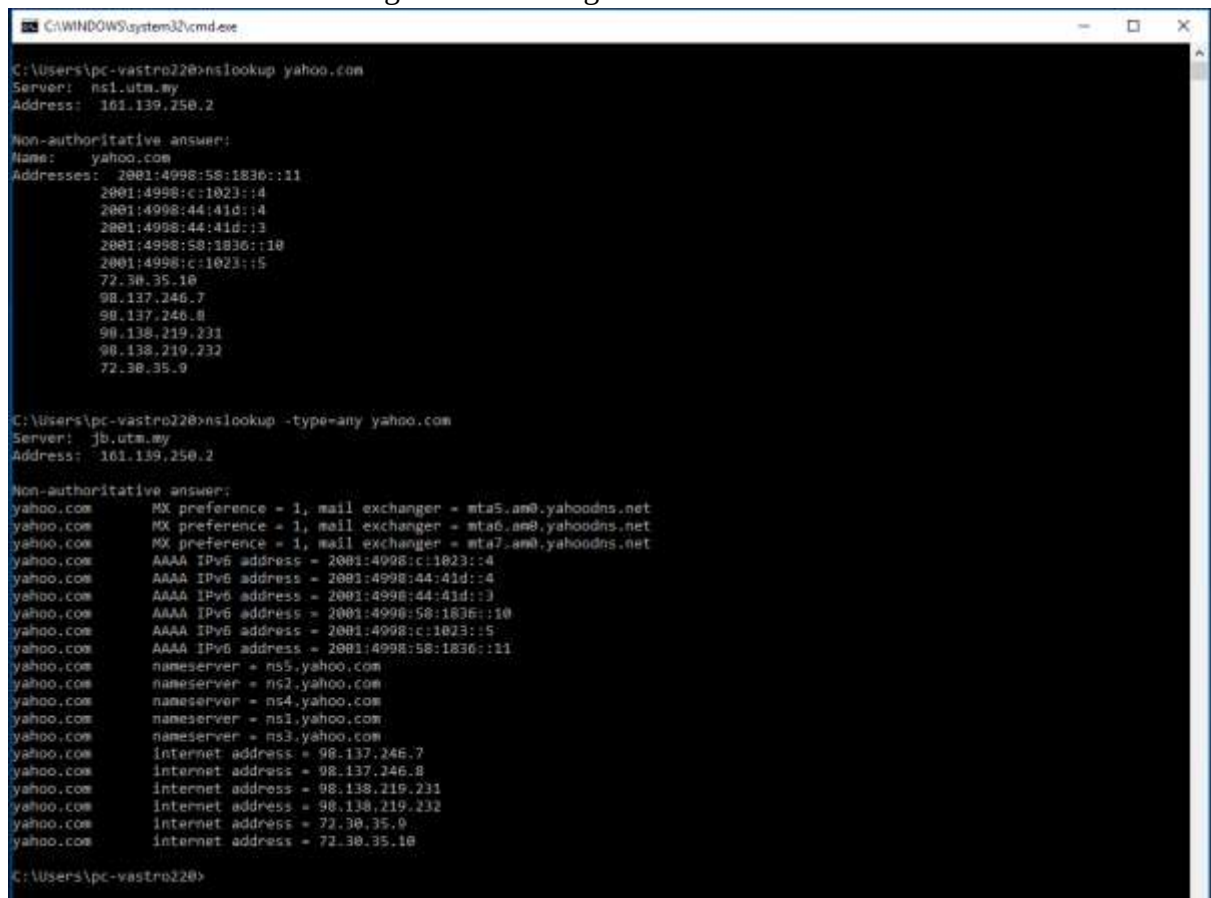
```
Server: NCSA/1.3.2\r\n
Last-modified: Tue, 23 Sep 20
Content-type: image/jpeg\r\n
Content-length: 15642\r\n
```


PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1830::11
          2001:4998:c:1023::4
          2001:4998:44:41d::4
          2001:4998:44:41d::3
          2001:4998:58:1830::10
          2001:4998:c:1023::5
          72.30.35.10
          98.137.246.7
          98.137.246.8
          98.138.219.231
          98.138.219.232
          72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1830::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1830::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
          172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

IPv6 addresses:

- 2001:e68:2:a5::356e
- 2001:e68:2:87::356e
- 2001:e68:2:84::356e
- 2001:e68:2:81::356e
- 2001:e68:2:88::356e

IPv4 address:

- 23.220.242.103

```
C:\Users\Nitro>nslookup www.microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
Name:     e13678.dscb.akamaiedge.net
Addresses: 2001:e68:2:a5::356e
           2001:e68:2:87::356e
           2001:e68:2:84::356e
           2001:e68:2:81::356e
           2001:e68:2:88::356e
           23.220.242.103
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

Non-authoritative answer:

microsoft.com nameserver = ns4-39.azure-dns.info
microsoft.com nameserver = ns1-39.azure-dns.com
microsoft.com nameserver = ns2-39.azure-dns.net
microsoft.com nameserver = ns3-39.azure-dns.org

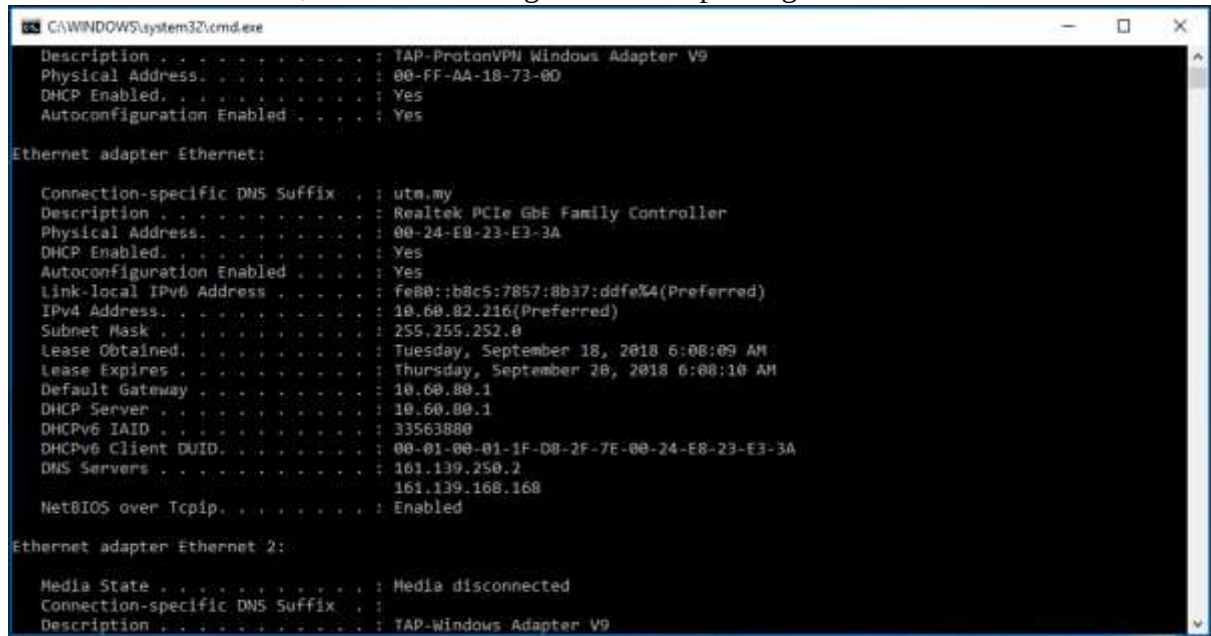
```
C:\Users\Nitro>nslookup -type=ns microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
microsoft.com nameserver = ns4-39.azure-dns.info
microsoft.com nameserver = ns1-39.azure-dns.com
microsoft.com nameserver = ns2-39.azure-dns.net
microsoft.com nameserver = ns3-39.azure-dns.org
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-00
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

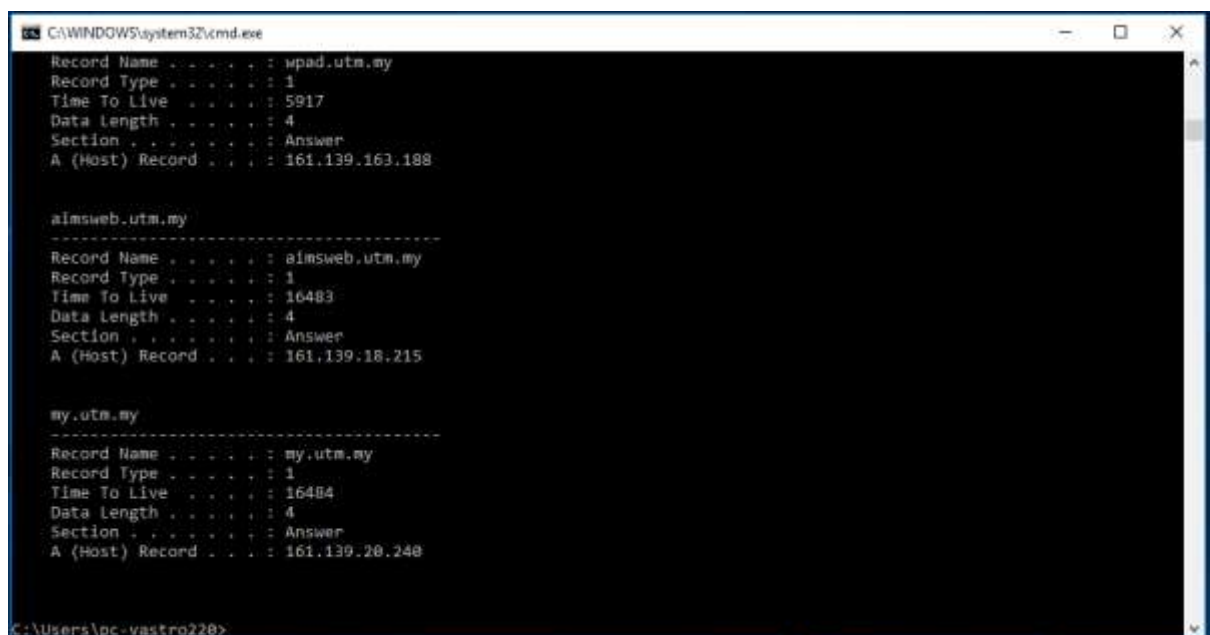
    Connection-specific DNS Suffix . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           : 161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds. Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro228>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns

```

C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>ipconfig /flushdns
Windows IP Configuration
Successfully flushed the DNS Resolver Cache.
C:\Users\pc-vastro220>

```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.
1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

	Time	Source	Destination	Protocol	Length	Info
	9 3.076689	128.238.29.23	128.238.38.160	DNS	104	Star
	8 3.075845	128.238.38.160	128.238.29.23	DNS	72	Star

Frame 8: Query

```

Frame 8: 72 bytes on wire (
Ethernet II, Src: IBM_10:60
Internet Protocol Version 4
User Datagram Protocol, Src
Domain Name System (query)

```

Frame 9: Response

```

Frame 9: 104 bytes on wire (83
Ethernet II, Src: Cisco_83:e4:
Internet Protocol Version 4, S
User Datagram Protocol, Src Po
Domain Name System (response)

```

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Query:

Destination port – 53

Response:

Source port – 53

```
Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits) on interface 0  
Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-MSRP-rot  
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23  
User Datagram Protocol, Src Port: 3163, Dst Port: 53  
Domain Name System (query)
```

```
Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on interface 0  
Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99  
Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160  
User Datagram Protocol, Src Port: 53, Dst Port: 3163  
Domain Name System (response)
```

3. To what IP address is the DNS query message sent? Add screenshots in your answer.

128.238.29.23

	Time	Source	Destination
9	3.076689	128.238.29.23	128.238.38.160
8	3.075845	128.238.38.160	128.238.29.23

4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type A, no answer.

```
▶ Frame 8: 72 bytes on wire (576 bits),
▶ Ethernet II, Src: IBM_10:60:99 (00:09:
▶ Internet Protocol Version 4, Src: 128.
▶ User Datagram Protocol, Src Port: 3163
▼ Domain Name System (query)
  Transaction ID: 0x006e
  ▶ Flags: 0x0100 Standard query
  Questions: 1
  Answer RRs: 0
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    ▶ www.ietf.org: type A, class IN
    [Response In: 9]
```

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

2 answers.

Answer 1:

```
▼ Answers
  ▼ www.ietf.org: type A, class IN, addr 132.151.6.75
    Name: www.ietf.org
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 1678 (27 minutes, 58 seconds)
    Data length: 4
    Address: 132.151.6.75
```

Answer 2:

```
▼ www.ietf.org: type A, class IN, addr 65.246.255.51
  Name: www.ietf.org
  Type: A (1) (Host Address)
  Class: IN (0x0001)
  Time to live: 1678 (27 minutes, 58 seconds)
  Data length: 4
  Address: 65.246.255.51
```

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

Yes

```
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-router
▼ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 132.151.6.75
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total length: 40
    Identification: 0x22a1 (8865)
  ▶ 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to live: 128
  Protocol: TCP (6)
  Header Checksum: 0xa5be [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 128.238.38.160
  Destination Address: 132.151.6.75
  [Stream index: 2]
```

Time	Source	Destination	Protocol	Length	Info
12 3.896461	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 #
17 3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376
19 3.148869	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376
21 3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376
22 3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376
18 3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 #
27 3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 #
35 3.222249	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=267
37 3.228509	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=267
42 3.238877	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [FIN, ACK] Seq=267
24 3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 #
38 3.192695	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [ACK] Seq=1 #
41 3.230667	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [ACK] Seq=263
44 3.246286	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [FIN, ACK] Seq=263
25 3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 #
49 3.286583	128.238.38.160	132.151.6.75	TCP	54	3372 → 80 [ACK] Seq=1 #
57 3.309371	128.238.38.160	132.151.6.75	TCP	54	3372 → 80 [ACK] Seq=264
60 3.311348	128.238.38.160	132.151.6.75	TCP	54	3373 → 80 [ACK] Seq=264

▼ Answers

```
▼ www.ietf.org: type A, class IN, addr 132.151.6.75
  Name: www.ietf.org
  Type: A (1) (Host Address)
  Class: IN (0x0001)
  Time to live: 1678 (27 minutes, 58 seconds)
  Data length: 4
  Address: 132.151.6.75
```


7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No, there is no new DNS queries before retrieving each image after the initial DNS query.

8	3.075045	128.238.38.160	128.238.29.25	DNS	72 Standard query 0x000e A www.ietf.org
9	3.076089	128.238.29.25	128.238.38.160	DNS	104 Standard query response 0x000e A www.ietf.org A 132.151.6.75 A 65.246.255.51
10	3.076479	128.238.38.160	132.151.6.75	TCP	62 3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62 80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54 3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429 GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60 80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434 80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.120693	132.151.6.75	128.238.38.160	TCP	1434 80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.120148	128.238.38.160	132.151.6.75	TCP	54 3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434 80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54 3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055 HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54 3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161067	128.238.38.160	132.151.6.75	TCP	54 3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60 80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62 3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62 3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	3.193649	132.151.6.75	128.238.38.160	TCP	62 80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
27	3.193726	128.238.38.160	132.151.6.75	TCP	54 3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
28	3.193998	128.238.38.160	132.151.6.75	HTTP	320 GET /images/ietflogo2e.gif HTTP/1.1
29	3.192665	132.151.6.75	128.238.38.160	TCP	62 80 → 3371 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
30	3.192695	128.238.38.160	132.151.6.75	TCP	54 3371 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
31	3.192869	128.238.38.160	132.151.6.75	HTTP	314 GET /images/blue.gif HTTP/1.1

- Open packet trace file dns-trace-2 for nslookup.
 - We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
 - Answer the following questions.
8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Query message - Destination Port: 53

Response message – Source Port: 53

▶	Frame 19: 71 bytes on wire (568 bits), 71 bytes captured
▶	Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst:
▶	Internet Protocol Version 4, Src: 128.238.38.160, Dst: 1
▼	User Datagram Protocol, Src Port: 3742, Dst Port: 53

▶	Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured
▶	Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst:
▶	Internet Protocol Version 4, Src: 128.238.29.22, Dst: 12
▶	User Datagram Protocol, Src Port: 53, Dst Port: 3742

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

128.238.29.22. No, it is not the IP address of my default local DNS server.

	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Sta
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Sta
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Sta
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Sta
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Sta
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Sta

```
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : dlinkrouter.local
Description . . . . . : Intel(R) Wi-Fi 6 AX203
Physical Address. . . . . : 4C-49-6C-5C-AB-DE
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2001:e68:5471:4a48:8e0b:b7a:9dfa:fd58(Preferred)
Temporary IPv6 Address. . . . . : 2001:e68:5471:4a48:e502:1696:d8ee:b61d(Preferred)
Link-local IPv6 Address . . . . . : fe80::6d83:dafc:2afd:b618%11(Preferred)
IPv4 Address. . . . . : 192.168.0.115(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, January 21, 2025 11:40:45 AM
Lease Expires . . . . . : Saturday, January 25, 2025 11:02:13 AM
Default Gateway . . . . . : fe80::e0e:76ff:fe68:b46d%11
                          192.168.0.1
DHCP Server . . . . . : 192.168.0.1
DHCPv6 IAID . . . . . : 105662828
DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-50-A5-7C-74-DD-43-C7-7D
DNS Servers . . . . . : 2001:e68::b:68
                          2001:e68::b:69
                          1.1.1.1
                          1.1.1.2
NetBIOS over Tcpip. . . . . : Enabled
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type A, no answers.

```
▼ Domain Name System (query)
  Transaction ID: 0x0003
  ► Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▼ Queries
    ► www.mit.edu: type A, class IN
    [Response In: 20]
```

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

1 answer

```
▼ Answers
  ▼ www.mit.edu: type A, class IN, addr 18.7.22.83
    Name: www.mit.edu
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 60 (1 minute)
    Data length: 4
    Address: 18.7.22.83
```

```
▼ Domain Name System (response)
  Transaction ID: 0x0003
  ▶ Flags: 0x8580 Standard query response, No error
  Questions: 1
  Answer RRs: 1
  Authority RRs: 3
  Additional RRs: 3
  ▶ Queries
  ▼ Answers
    ▼ www.mit.edu: type A, class IN, addr 18.7.22.83
      Name: www.mit.edu
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 60 (1 minute)
      Data length: 4
      Address: 18.7.22.83
```